



CS 502: Compiler Design

Course Code	CS 502
Course Name	Compiler Design
L-T-P-C	3-0-2-4
Prerequisites	CS 202: Data Structures and Algorithms; CS304: Formal Languages and Automata Theory
Intended for	BTech third and final year CSE and MS/PhD
Distribution	Elective for third and final year BTech CSE/EE, MS, PhD
Approval	33rd BoA (Jan 2020); 24th Senate (Feb 2020)

Course Contents

- **Introduction and lexical analysis** Introduction to language translators. Stages of compilation. Lexical analyzers: token specification and recognition. [4 Lectures]
- **Parsing** Overview of context-free grammars. Parse trees and derivations. Left recursion and left factoring. Top-down and bottom-up parsing. [8 Lectures]
- **Semantic analysis** Syntax-directed translation. Various intermediate representations. Intermediate code generation. Type checking. [9 Lectures]
- **Runtime environments:** Activation records. Heap management. Garbage collection. [4 Lectures]
- **Code generation and optimization:** Register allocation. Instruction selection and scheduling. Control-flow graphs. Data-flow analysis. Peephole optimizations. [9 Lectures]
- **Advanced topics:** Loop optimizations. Call-graph construction. Machine learning in compiler design. Just-in-time compilers. [8 Lectures]

Labs

Learning the art of compiler design involves a nice mix of theory and practice. The course consists of at least four programming assignments from the following set (assignments will be rotated in every offering to contain plagiarism):

- Learning to use a lexer/parser generator such as Flex/Bison/JavaCC
- Intermediate code generation
- Type checking
- One forward/backward dataflow analysis
- Register allocation
- Assembly code generation

Text Books

1. Alfred V. Aho, Monica Lam, Ravi Sethi and Jeffrey D. Ullman, **Compilers: Principles, Techniques, and Tools**, 2nd Edition, Pearson Education, 2007.
2. Andrew W. Appel, Jens Palsberg, **Modern Compiler Implementation in Java**, 2nd Edition, Cambridge University Press, 2002.

References

1. Keith D. Cooper and Linda Torczon, **Engineering a Compiler**, 2nd Edition, Morgan Kaufmann, 2011.
2. Y. N. Srikant and Priti Shankar, **The Compiler Design Handbook**, 2nd Edition, CRC Press, 2007.